DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
PRE BOARD EXAMINATION

CLASS: X **FULL MARKS: 80**
SUBJECT: PHYSICS [SET A] **TIME: 2 HOURS**

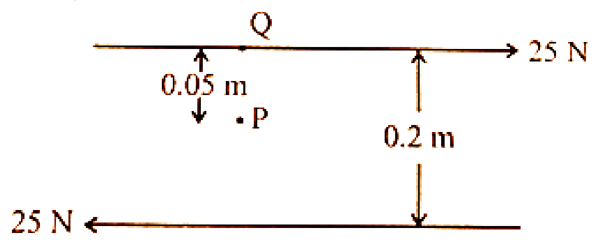
Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.
Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This paper consists of eight printed pages

SECTION A

Question 1 **[15 × 1 = 15]**

Choose the correct answers to the questions from the given options:
(Do not copy the question, write the correct answers only.)

(i) From the given figure, calculate moment of force about (a) P and (b) Q



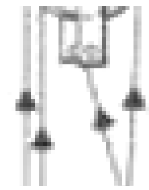
(a) 10Nm clockwise, 5 Nm anticlockwise
(b) 5 Nm clockwise, 10 Nm anticlockwise
(c) 10 Nm clockwise, 10 Nm anticlockwise
(d) 5 Nm clockwise, 5 Nm clockwise

(ii) In a simple microscope, increasing the focal length of the lens
(a) increases magnifying power
(b) decreases magnifying power
(c) increases image distance
(d) has no effect

(iii) An aeroplane is flying at an altitude of 10000 m at a speed of 300 km/h. The aeroplane at this height has:
(a) zero kinetic and potential energy
(b) only potential energy
(c) only kinetic energy
(d) both kinetic and potential energy

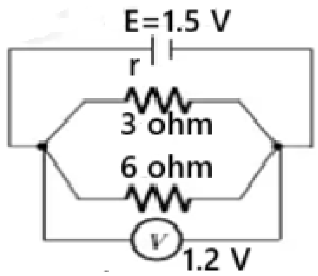
(iv) The amplitude of forced vibrations is generally ___________than the amplitude of applied external force.
(a) more **(b) equal to**
(c) less **(d) both (a) and (b)**

(v) The adjacent diagram shows a portion of a block and tackle system. The direction of effort is not specified. What could be the probable mechanical advantage of the system?



(a) 4 (b) 5 (c) 3 (d) both (a) and (b)

(vi) Study the circuit diagram in figure and hence, calculate the internal resistance of the cell.



(a) 0.3 ohm (b) 2.8 ohm
(c) 0.5 ohm (d) 3.0 ohm

(vii) In a parallel circuit
(a) the equivalent resistance of all resistors is more than any of the individual resistors
(b) potential difference across each resistor is different
(c) current flowing through all resistors is same
(d) the equivalent resistance of all resistors is less than any of the individual resistors

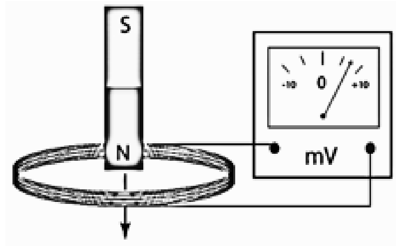
(viii) The focal length of a concave lens is 20 cm. What will be its image distance if the object is placed at 20 cm?
(a) 10 cm
(b) 20 cm
(c) anywhere between 10 cm and 20 cm.
(d) at infinity

(ix) Assertion (A) : Power of a convex lens is positive and that of a concave lens is negative.
Reason (R) : A lens of short focal length deviates the rays more, while a thin lens deviates the rays less.
(a) both (A) and (R) are true and (R) is the correct explanation of (A).
(b) both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is false but (R) is true.
(d) (A) is true but (R) is false.

(x) When 100 g of water at 0°C is mixed with 100 g of ice at 0°C, the temperature of the mixture will be:
(a) 0°C
(b) below 0°C
(c) between 0°C and 10°C
(d) depends on the latent heat of ice

(xi) In a large empty room, a teacher's voice sounds prolonged and unclear. This is mainly due to:
(a) echo formation from distant walls
(b) resonance between the walls and sound waves
(c) reverberation caused by multiple reflections
(d) damping of sound due to air resistance

(xii) **A magnet is dropped through a coil connected to a millivoltmeter as shown. The voltmeter needle moves to the right showing a positive reading.**



Which of these will make the voltmeter reading smaller but still positive?
(a) drop the south pole through first.
(b) add more turns to the coil.
(c) move the magnet slowly through the coil.
(d) use a stronger magnet.

(xiii) **How many alpha and beta particles are emitted when Uranium $_{92}U^{238}$ decays to lead $_{82}Pb^{206}$?**
(a) 3 alpha and 5 beta particles
(b) 6 alpha and 2 beta particles
(c) 4 alpha and 5 beta particles
(d) 8 alpha and 6 beta particles

(xiv) **A mass is suspended from a spring and set into vertical oscillation. How can the frequency of oscillation to be decreased?**
(a) By increasing the mass
(b) By decreasing the mass
(c) By using a stiffer spring
(d) Only A and C are correct

(xv) **A student changed the material of the calorimeter from copper to another metal. He observed that for the same amount of heat supplied, the temperature rise of the water was slightly less than before. This suggests that the specific heat capacity of the new calorimeter material is:**
(a) greater than that of copper
(b) less than that of copper
(c) equal to that of copper
(d) negligible compared to copper.

Question 2

(i) **Complete the following by choosing the correct answers from the bracket:** **[6]**

(a)When a spring is pulled, then the work done by external force __________________ (positive/negative/zero).
(b) 10^{-10} Nm is equal to __________dyne cm. [0.001 /10/ 10^{-4}].
(c) The _____________(frequency/ speed/ intensity) of a wave is determined by the properties of the medium through which it travels.
(d) A nuclear reactor uses __________ (cadmium/ graphite/ boron) as a solid moderator to slow down fast neutrons.
(e)The image is formed _____________________[at F_2 /at $2F_2$ /beyond $2F_2$] in a terrestrial telescope.

(f) The boiling point of water ____________ [decreases / increases/remains same] with increase in pressure.

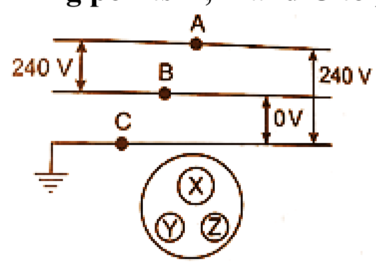
(ii) Match the classes of levers as given in Column A with that of Column B [2]

Column A	Column B
Wrench	• Class I
Stapler	• Class II
	• Class III

(iii) (a) Name the radiations used for checking welding in industries. [2]
(b) Mention one property of such radiations not similar to visible light.

Question 3

(i) Redraw the diagram by linking points A, B and C to points X, Y and Z on the socket. [2]



(ii) (a) Explain why the electric power transmission lines use very high voltages for long-distance transmission.
(b) State one reason why we prefer a ring system as a house wiring system. [2]

(iii) (a) Draw a graph showing the variation of terminal potential difference with current for a cell.
(b) What will be the slope of the above graph? [2]

(iv) A solid of mass 60 g at 100°C is placed in 150 g of water at 20°C. The final temperature is 25°C. Calculate the specific heat capacity of the solid. [sp. heat capacity of water is 4.2 $Jg^{-1}K^{-1}$] [2]

(v) (a) Name the principle on which A.C generator works.
(b) Mention any one way to increase the speed of rotation of the coil in a DC motor. [2]

(vi) (a) When does the nucleus on an atom become radioactive?
(b) Give an example of tracers. [2]

(vii) A special task force is conducting a rescue operation in the high-altitude Himalayas. The surroundings have intense sunlight, very low air temperature, and thick snow cover. The team uses a medical kit that must be kept at a stable temperature and a portable stove for melting snow to obtain drinking water. They have two liquids available for temperature stabilization:
liquid A: Specific Heat Capacity = 4200 J kg^{-1} K^{-1}
liquid B: Specific Heat Capacity = 800 J kg^{-1} K^{-1}
(a) Which liquid should they use as a thermal reservoir for the medical kit? Give a reason.
(b) At night, the air just above a frozen glacial lake remains slightly warmer for longer than the air above dry ground. Explain this observation in terms of the process of freezing.
(c) Why is melting of snow on a stove slower at high altitude even though sunlight is intense? [3]

SECTION B
(Attempt any four questions)

Question 4 **[3+3+4]**

(i) **A swimmer under water looks up at an aeroplane flying overhead.**

(a) How will the aeroplane appear to him-higher or lower than the actual height?

(b) When a pilot flying a plane looks down at a swimmer in a pool, how will the swimmer appears to the pilot?

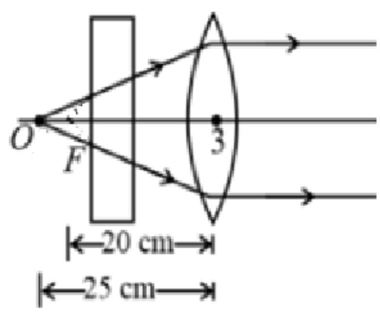
(c) What is the angle of deviation suffered by a light ray in case of an erecting prism?

(ii) **A point object is placed at a distance of 25*cm* from a convex lens of focal length 20*cm*. If a glass slab of thickness *t* and refractive index 1.5 is inserted between the lens and object. Observe the diagram carefully and answer the questions:**

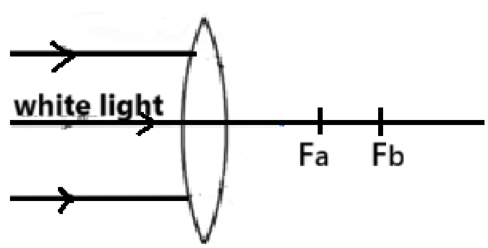
(a) The rays coming out from the lens are parallel even though the object O is not placed at the focus. Explain in brief.

(b) Where will the final image be formed?

(c) How will the apparent position of the object change if a glass slab of greater thickness is taken?



(iii) **(a) A beam of white light is incident on a convex lens, as shown in the figure below.**



Identify the focus of the red colour from the diagram. [ray diagram not needed]. Also give reasons for your selection.

(b) 1. Name the physical phenomenon responsible for the working of a megaphone.

2. State the condition on the size of the reflecting surface for the megaphone to work effectively.

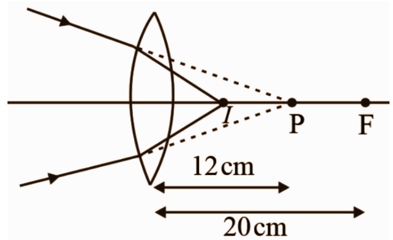
Question 5 **[3+3+4]**

(i) **(a) Name the phenomenon that occurs when white light is incident on a prism.**

(b) Does deviation occur on both surfaces of the prism? Why do the colours get further

separated as they emerge through the second surface?

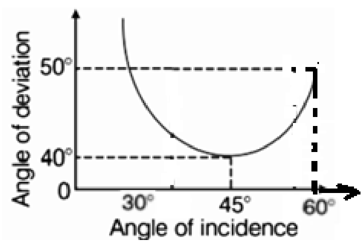
(ii) A beam of light converges at a point P. Now a lens is placed in the path of the convergent beam 12 cm from P.



(a) At what point does the beam converge if the lens is a convex lens of focal length 20 cm.

(b) Calculate its magnification (upto one decimal point)

(iii) (a) The diagram below shows the plotting of angle of deviation with angle of incidence.



1. Determine the angle of the prism from the given graph.
2. Using the graph, identify the condition of the prism when the deviation is minimum.

(b) If the same glass prism is in water, justify whether the deviation exceeds or be less than 40°?

Question 6 **[3+3+4]**

(i) For each of the following situations, state whether the work done by gravity is positive, negative or zero.

(a) a polar satellite orbiting the earth.

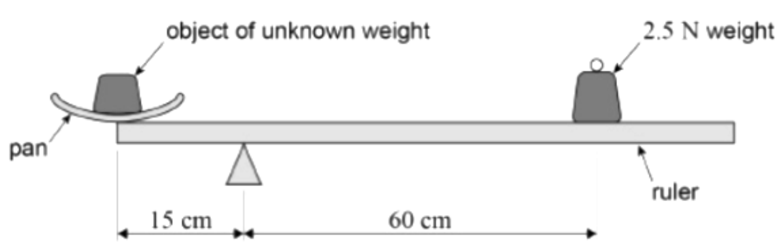
(b) a boy climbing up a ladder.

(c) a vehicle in neutral gear moving down an inclined plane.

(ii) The diagram shows a uniform metre ruler of weight 1.5 N pivoted 15 cm from one end for use as a simple balance. A scale pan of weight 0.5 N is placed at the end of the ruler and an object of unknown weight is placed in the pan. The ruler attains to a steady horizontal position when a weight of 2.5 N is added at a distance of 60 cm from the pivot as shown.

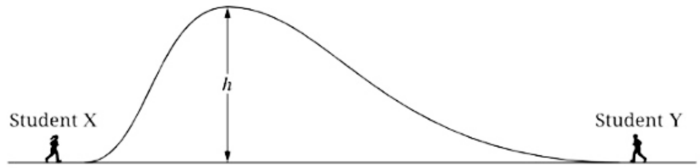
(a) What is the weight of the object?

(b) How will the weight at the other end change if the pivot is moved 5 cm toward the scale pan?



(iii) (a) Two students of equal mass run to the top of a hill from opposite sides, as shown in the figure. Both students reach the same height, but student Y takes half the time that student X takes to reach the top.

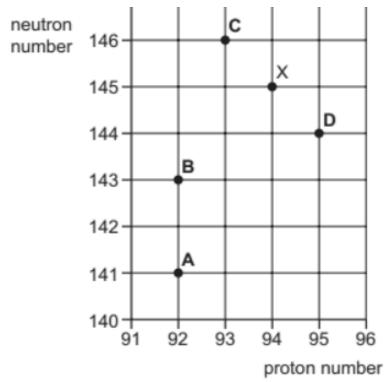
1. How does the rate of work done by the Earth's gravitational force on student X compare to that on student Y while they run up the hill?
2. Find the ratio of the power of the two students.



(b) Name the force that is used by the washing machine to wash the clothes. Is the force real or pseudo?

Question 7 **[3+3+4]**

(i) **(a) The figure shows a part of a chart of nuclides where neutron number is plotted against the proton number. An unstable nuclide X decays by emitting an alpha particle. Find the nuclide formed by the decay of nuclide X from the graph?**



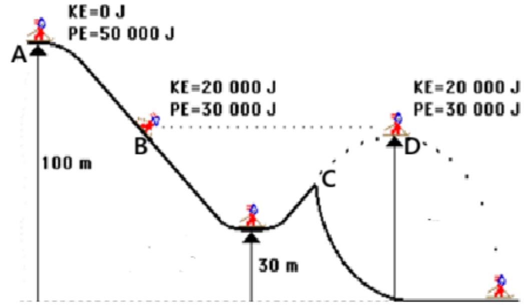
(b) State two differences between a forced vibration and resonance.

(ii) **A person standing between two vertical cliffs and is 640 m away from the nearest cliff. He shouted and heard the first echo after 4 s and the second echo after further 3s. Calculate the (a) speed of sound in air and**
(b) the distance between the cliffs.

(iii) **During a school science fair, Riya demonstrates an experiment using a tuning fork and a resonance tube partly filled with water. She strikes the tuning fork of frequency 512 Hz and holds it above the open end of the tube. By adjusting the water level, she observes that a loud sound is heard when the air column length is about 16 cm and again at 48 cm.**
(a) Explain why the sound is louder at these specific lengths.
(b) Which phenomenon is being demonstrated?
(c) If the tuning fork is replaced by one of lower frequency, how will the resonant lengths change? Give a reason.

Question 8 **[3+3+4]**

(i) **A skier starts from rest from point A which is at a height of 100 m and reaches at point C from where he jumps in the air.**



Observe the diagram carefully and write the following answers:

(a) What is the height of point D?

(b) Find the speed of the body at D. [given g = 10 N/kg]

(ii) (a) Distinguish between nuclear fusion and nuclear fission.[2 points]

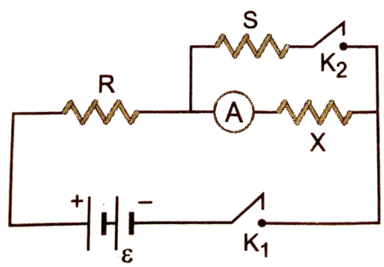
(b) Why can't nuclear fusion be performed in a nuclear reactor?

(iii) A 2 Ω (R) resistor is in series across a parallel combination of 4 Ω (S) and 12 Ω (X) resistors as shown. The current through R is 2.4 A.

(a) Determine the emf of the cell.

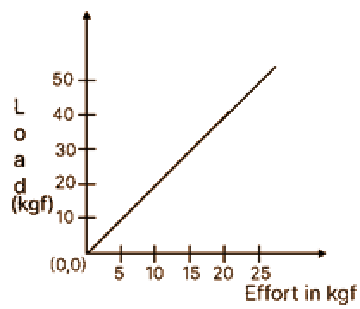
(b) Find the reading of ammeter A when K_2 is closed.

(c) Show whether the above reading increases or decreases when the key K_2 is open.



Question 9 **[3+3+4]**

(i) The graph shows load against effort for a lever with load and effort on the either side of the fulcrum.



(a) Calculate the mechanical advantage from the above graph.

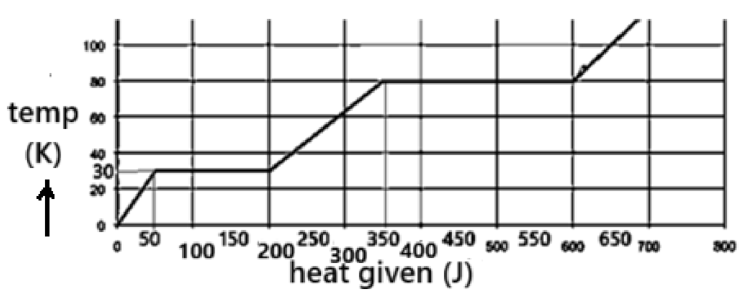
(b) Which class does this lever belong to?

(c) How did you arrive at this conclusion?

(ii) A substance is in the form of a solid at 0°C. The amount of heat added to this substance and the temperature of the substance are plotted on the graph. If the specific heat capacity of the solid substance is 150 J/kg°C, find from the graph:

(a) the mass of the substance

(b) the specific latent heat of fusion of the substance in the liquid state.



(iii) (a) Draw a labelled diagram of a simple D.C motor.

(b) Why the primary coil is thicker than that of the secondary coil in the step up transformers?